

Chahak Jain

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C++ SOFTWARE ENGINEER

LOW LATENCY SYSTEMS — MULTITHREADING — TCP/UDP — SOCKET PROGRAMMING

Performance-driven Software Engineer with 2+ years of experience building high-throughput, multi-threaded backend systems using C++. Specialized in socket programming, asynchronous I/O (IOCP), and performance optimization for real-time applications. Proven track record of improving system throughput by 3x and reducing load latency by over 90%. Strong expertise in system-level debugging, memory optimization, and concurrent system design.

TECHNICAL SKILLS

Languages: C/C++, SQL, JavaScript, Rust(basic), HTML/CSS

Backend & Systems: Socket Programming(TCP/UDP), Multi-threading, Asynchronous I/O, Network Protocols, Concurrent Systems, IOCP & Epoll

Core Concepts: Data Structures, STL, OOP, Memory Management, Operating Systems, Low-Latency Programming, Performance Optimization, Distributed Systems, Database Management, Mutex, Atomic, Thread, Context Switching

Frameworks & Tools: Windows Systems Programming, Linux, Git, Visual Studio, VS Code, Docker, Eclipse

EXPERIENCE

Software Engineer

January 2025 – Present

Synapsewave Innovations

Mumbai, India

- Developed and optimized latency-sensitive components of institutional trading platform handling real-time market data feeds and order execution using C++ and multi-threading.
- Performed performance profiling using Visual Studio Profiler and system-level debugging to identify bottlenecks, reducing application hangs by 90% and improving system responsiveness.
- Identified and resolved functional, logical, and crash issues by **analyzing application logs and crash dumps**, improving system reliability and user trust.

Software Engineer Intern → Software Engineer

February 2023 – January 2025

63 Moons (Formerly Financial Technology)

Mumbai, India

- Designed and implemented a **multi-threaded data processing pipeline using C++**, increasing throughput from 200 to 600+ messages/sec (3x improvement).
- Re-engineered the communication architecture by separating interactive messages, broadcast messages, and UI operations into dedicated threads, improving responsiveness and stability.
- Reduced application view load time from 60 seconds to 5 seconds by introducing optimized caching mechanisms using **arrays, linked lists, and hash maps for efficient data storage, search, and deletion**.
- Worked to provide modular functionality and enhance system stability by integrating and debugging both **static and dynamic DLLs**.

PROJECTS

Multi-Threaded Chat Client | C++, TCP/IP, Multithreading, IOCP

- Built a high-performance multi-threaded chat client using C++ and TCP sockets supporting concurrent client communication.
- Implemented asynchronous I/O using Windows IOCP to handle multiple connections efficiently with minimal thread overhead.
- Designed thread-safe communication using synchronization primitives, ensuring reliable message delivery.
- Optimized network and threading model to reduce blocking and improve responsiveness under concurrent workloads.

EDUCATION

Rajasthan Technical University

Bachelor of Technology in Computer Science

8.53 CGPA

2019 – 2023

Aadarsh Vidhyalay

Senior Secondary Education

84.7%

2018 – 2019

Modern Academy

Secondary Education

96.7%

2016 – 2017